

Metadata files (icons, social images, SEO)

Live demo: `/metadata` (+ `/sitemap.xml`, `/robots.txt`) **Code:** the files live at the top of `src/app/`; this page indexes them.

What it is

Drop specially-named files in `app/` and Next.js automatically wires up the `<head>` tags or serves the file — no manual `<link>` / `<meta>` needed. There are two flavors: **static** files (e.g. `favicon.ico`) and **generated** files (`.tsx` / `.ts` that produce the asset in code).

The files in this lab (all at `app/`)

File	Produces	Open
<code>favicon.ico</code>	Tab icon	<code>/favicon.ico</code>
<code>icon.tsx</code>	<code><link rel="icon"></code> (generated)	<code>/icon</code>
<code>apple-icon.tsx</code>	<code><link rel="apple-touch-icon"></code>	<code>/apple-icon</code>
<code>opengraph-image.tsx</code>	<code>og:image</code> social preview	<code>/opengraph-image</code>
<code>twitter-image.tsx</code>	<code>twitter:image</code>	<code>/twitter-image</code>
<code>sitemap.ts</code>	<code>/sitemap.xml</code>	<code>/sitemap.xml</code>
<code>robots.ts</code>	<code>/robots.txt</code>	<code>/robots.txt</code>

Generated images (`ImageResponse`)

`icon.tsx`, `opengraph-image.tsx`, etc. export a default function returning `new ImageResponse(<jsx>, { width, height })` plus route-segment config like `size`, `contentType`, and `alt`. Next runs them and injects the right tags.

Code-based SEO

```
// sitemap.ts → /sitemap.xml
export default function sitemap(): MetadataRoute.Sitemap { return [ ... ]; }

// robots.ts → /robots.txt
export default function robots(): MetadataRoute.Robots { return { rules: {...}, sitemap: "..."; } }
```

Also: the Metadata API

Beyond files, you export a `metadata` object (or `generateMetadata()`) from a `layout` / `page` for `<title>`, `description`, etc. See the root `app/layout.tsx` (`title.template`, `metadataBase`) and `blog/[slug]` (`generateMetadata`).

Interview Q&A

- **Q: How do you add a favicon/OG image in the App Router?** Put `favicon.ico` / `opengraph-image.(png|tsx)` in `app/`; Next injects the tags automatically. Use a `.tsx` file with `ImageResponse` to generate them.

- **Q: How do you generate a sitemap or robots file?** Export a default function from `app/sitemap.ts` / `app/robots.ts` ; Next serves `/sitemap.xml` and `/robots.txt` .
- **Q: Static vs generated metadata files?** Static = a real asset file (e.g. `icon.png`). Generated = a `.ts` / `.tsx` file whose code produces the asset at build/request time.